Exploring Challenges and Good Practices in UX Designer-Developer Collaboration: A Systematic Literature Review

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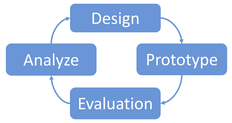
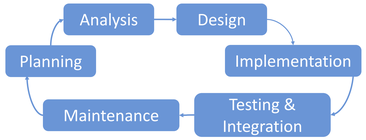
Introduction

Software development hinges on effective collaboration between Software Development Engineers (SDEs) and User Experience Designers (UXDs). Few previous studies cover the entire process of SE and UX collaboration, and most of them focus only on agile development.



UX Design's Life Cycle [2]



SE's Software Development Life Cycle (SDLC) [1]

RQ1: What are the challenges of UX and SE collaboration?
RQ2: What are the potential solutions to overcome the challenge?
RQ3: What are the existing tooling supports and gaps?

Review Method

Systematic Literature Review (keyword search among 5 databases including IEEE, ACM, Springer, Welly, and Engineering Village + two rounds of check for inclusion and exclusion)

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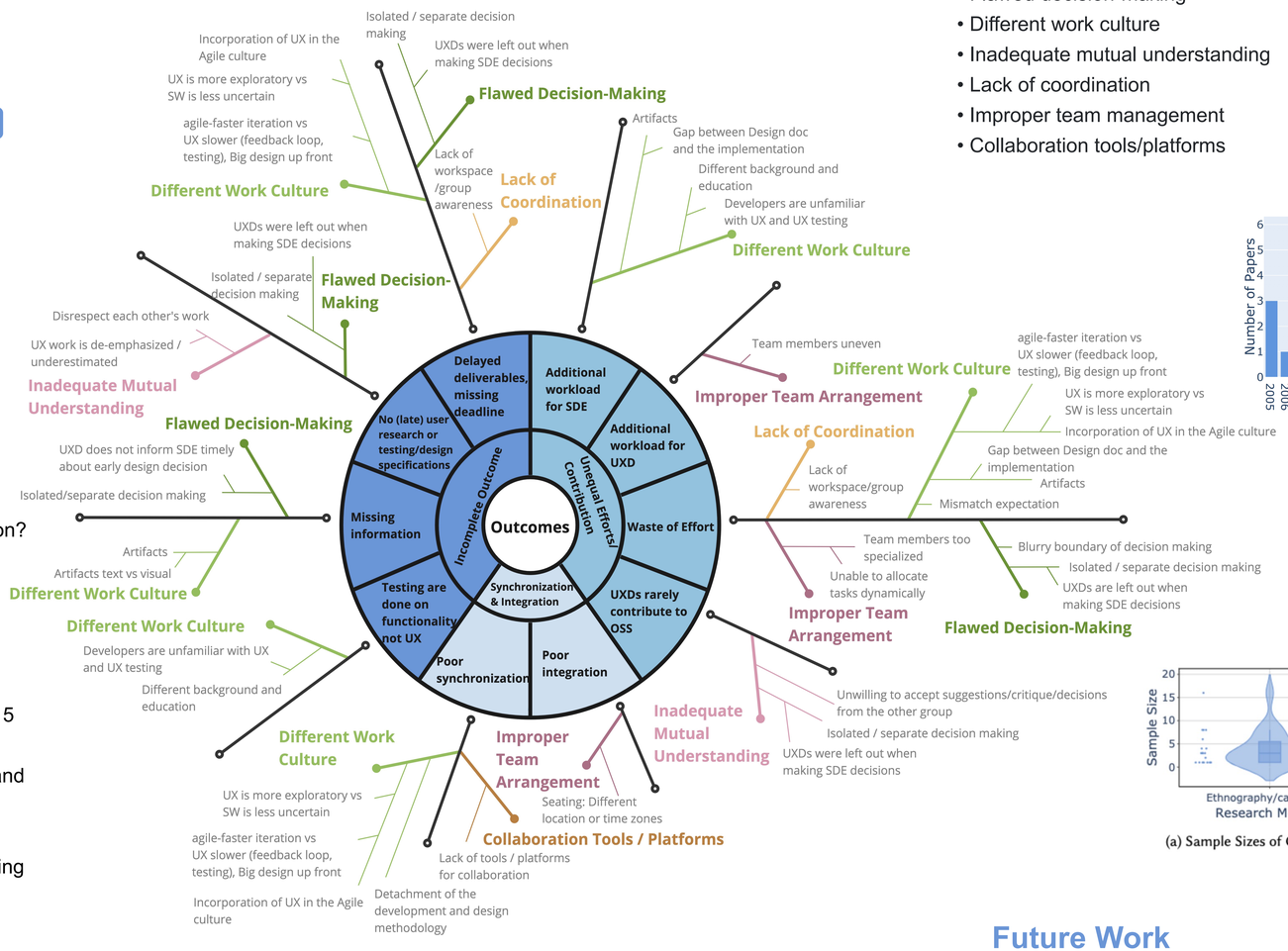
Grey Literature Review (4 popular online forums including UX Stack Exchange, Reddit UX, Reddit SE, and Stack Overflow + the VS Code project hosted on GitHub)

Results



Challenges

- Flawed decision-making
- Different work culture
- Inadequate mutual understanding
- Lack of coordination
- Improper team management
- Collaboration tools/platforms

Good Practices

- Designers start ahead
- Share artefacts
- Enhance communication
  - Explicit expectation
  - Frequent communication
  - Friendly relationship
- Promote UX
- Encourage inclusivity
  - Involve developers in UX
  - Involve designers in SE
- Adopt background & culture
- Improve tools & methods

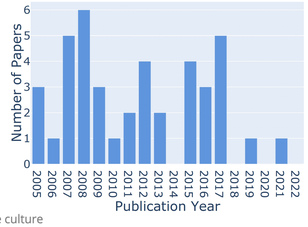


Fig. 1. Publication Years

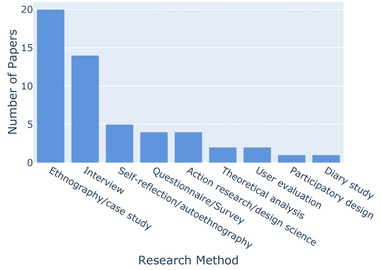
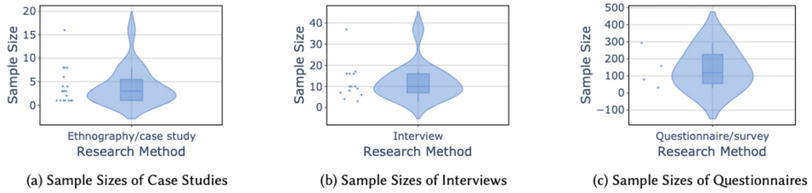


Fig. 2. Research Methods Used



(a) Sample Sizes of Case Studies (b) Sample Sizes of Interviews (c) Sample Sizes of Questionnaires

Fig. 3. Sample Sizes of Different Research Methods

Future Work

Our findings have broad applicability, extending beyond agile development to traditional and open-source software development as well. The proposed good practices can also serve as a guide for future studies, facilitating the creation of specific collaboration tools for SDEs and UXDs.

References

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